The Inclusion–Exclusion Principle

The sieve method

1. Two finite sets

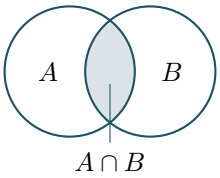
Let A and B be finite sets. The *sum rule* says that, if $A \cap B = \varnothing$, then

$$|A \cup B| = |A| + |B|.$$

More generally,

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|.} \tag{1}$$

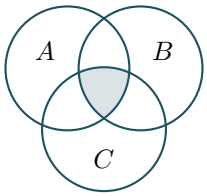
Indeed, in $|A| + |B|$, every element of $A \cap B$ is counted twice. Subtracting $|A \cap B|$ leaves every element of the union counted once.



2. Three finite sets

For finite sets A, B, C, the corresponding formula is

$$\begin{aligned} |A \cup B \cup C| = {} & |A| + |B| + |C| \\ & - |A \cap B| - |B \cap C| - |C \cap A| \\ & + |A \cap B \cap C|. \end{aligned} \tag{2}$$



Exercise. Why is (2) true? Check the contribution of an element belonging to exactly one, exactly two, or all three sets.

3. The general formula

Let $A_1, \dots, A_n$ be finite sets, where $n \geq 1$. Inclusion–exclusion alternately adds and subtracts the sizes of their intersections:

$$\begin{aligned}\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{1 \leq i_1 < i_2 \leq n} |A_{i_1} \cap A_{i_2}| \\ + \sum_{1 \leq i_1 < i_2 < i_3 \leq n} |A_{i_1} \cap A_{i_2} \cap A_{i_3}| - \cdots \\ + (-1)^{n-1} |A_1 \cap \cdots \cap A_n|.\end{aligned} \tag{3}$$

To write this more compactly, put $[n] = \{1, 2, \dots, n\}$ and let $\binom{X}{k}$ denote the *set of all k-element subsets* of X. Then

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{k=1}^{n} (-1)^{k-1} \sum_{I \in \binom{[n]}{k}} \left|\bigcap_{i \in I} A_i\right|. \tag{4}$$

Equivalently, using just one sum,

$$\boxed{\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|-1} \left|\bigcap_{i \in I} A_i\right|.} \tag{5}$$

4. Proof by counting contributions

Fix an element $x \in A_1 \cup \cdots \cup A_n$. Suppose x belongs to exactly r of the sets. After relabelling, we may assume these are $A_1, \dots, A_r$.

On the left-hand side of (5), x is counted once. On the right-hand side, it belongs to exactly $\binom{r}{k}$ of the intersections indexed by k-element subsets. Thus its total contribution is

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r-1} \binom{r}{r}.$$

This expression equals 1, since the binomial theorem gives

$$0 = (1-1)^r = 1 + \sum_{k=1}^{r} (-1)^k \binom{r}{k}.$$

Every element of the union therefore contributes exactly 1 to both sides. Elements outside the union contribute 0. This proves the formula. □

Exercise (inductive proof). Give another proof by induction on n, using the two-set identity (1) as the base case.

5. Proof using characteristic functions

Let X be a finite set containing $A_1, \dots, A_n$. For each A_i, define its *characteristic function* $f_i : X \to \{0, 1\}$ by

$$f_i(x) = \begin{cases} 1, & x \in A_i, \\ 0, & x \notin A_i. \end{cases}$$

The function $1 - f_i$ is the characteristic function of $X \setminus A_i$. Consequently,

$$F(x) := \prod_{i=1}^{n} (1 - f_i(x))$$

is the characteristic function of $X \setminus \bigcup_{i=1}^{n} A_i$: the product is 1 precisely when x belongs to none of the sets.

Expanding the product gives

$$F(x) = \sum_{I \subseteq [n]} (-1)^{|I|} \prod_{i \in I} f_i(x). \tag{6}$$

For every $I \subseteq [n]$, the product $\prod_{i \in I} f_i(x)$ is the characteristic function of $\bigcap_{i \in I} A_i$. We use the conventions

$$\prod_{i \in \varnothing} f_i(x) = 1, \qquad \bigcap_{i \in \varnothing} A_i = X.$$

For any $S \subseteq X$, its cardinality is the sum of its characteristic function:

$$|S| = \sum_{x \in X} \mathbf{1}_S(x).$$

Summing (6) over $x \in X$ and interchanging the finite sums, we obtain

$$\begin{aligned} \left| X \setminus \bigcup_{i=1}^{n} A_i \right| &= \sum_{x \in X} F(x) \\ &= \sum_{I \subseteq [n]} (-1)^{|I|} \sum_{x \in X} \prod_{i \in I} f_i(x) \\ &= \sum_{I \subseteq [n]} (-1)^{|I|} \left| \bigcap_{i \in I} A_i \right|. \end{aligned}$$

This is the *complement form* of inclusion–exclusion. The term $I = \varnothing$ is $|X|$, so subtracting the identity from $|X|$ yields

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|-1} \left| \bigcap_{i \in I} A_i \right|,$$

as required. □

The following sections collect useful versions of the principle, together with short proofs and standard applications.

6. Avoidance, prescribed membership, and weights

Throughout, $A_1, \dots, A_n \subseteq X$, with X finite. Write $A_I = \bigcap_{i \in I} A_i$ and $A_\varnothing = X$.

Avoiding all forbidden properties. If A_i consists of objects having forbidden property i, then the number having none of the forbidden properties is

$$\boxed{\left| X \setminus \bigcup_{i=1}^{n} A_i \right| = \sum_{I \subseteq [n]} (-1)^{|I|} |A_I|.} \tag{7}$$

This is often the most convenient form of PIE: count all objects first, then subtract those violating at least one condition.

Requiring some properties and forbidding others. For disjoint $P, Q \subseteq [n]$, the number of objects belonging to every A_i with $i \in P$ and to no A_j with $j \in Q$ is

$$\left| A_P \setminus \bigcup_{j \in Q} A_j \right| = \sum_{J \subseteq Q} (-1)^{|J|} |A_{P \cup J}|. \tag{8}$$

Apply the avoidance formula inside the universe A_P. There is no restriction on indices outside $P \cup Q$. Taking $Q = [n] \setminus P$ counts objects whose membership set is *exactly* P.

For example, the number lying in A_1, A_2, but in neither A_3 nor A_4, is

$$|A_{\{1,2\}}| - |A_{\{1,2,3\}}| - |A_{\{1,2,4\}}| + |A_{\{1,2,3,4\}}|.$$

The dual form: intersections from unions. For $n \geq 1$,

$$\left| \bigcap_{i=1}^{n} A_i \right| = \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|-1} \left| \bigcup_{i \in I} A_i \right|. \tag{9}$$

To prove this, apply PIE to the complements $X \setminus A_i$ and use De Morgan's laws. The coefficients of $|X|$ cancel because $\sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|-1} = 1$. Use this form when unions are easier to count than intersections.

Weighted PIE. Assign each $x \in X$ a weight $w(x)$ and put $W(B) = \sum_{x \in B} w(x)$. Then

$$W\left(X \setminus \bigcup_{i=1}^{n} A_i \right) = \sum_{I \subseteq [n]} (-1)^{|I|} W(A_I). \tag{10}$$

Multiply the indicator identity by $w(x)$ and sum over x. The weights may be real or complex numbers, or elements of any abelian group. In particular, polynomial weights give identities of generating functions. All the exact counting identities in this note have the same weighted versions; inequalities require nonnegative real weights.

7. Exactly or at least a given number of properties

Let $m(x) = |\{i : x \in A_i\}|$ and define

$$N_r = |\{x \in X : m(x) = r\}| \quad (0 \le r \le n), \qquad S_k = \sum_{\substack{I \subseteq [n] \\ |I|=k}} |A_I|, \quad S_0 = |X|.$$

Thus S_k counts pairs (x, I) with $|I| = k$ and $x \in A_I$; an object in j sets contributes $\binom{j}{k}$. Empty sums give $S_k = 0$ for $k > n$.

The sieve polynomial. All the multiplicity counts are contained in one identity:

$$\boxed{P(z) := \sum_{r=0}^{n} N_r z^r = \sum_{k=0}^{n} S_k (z-1)^k.} \tag{11}$$

Indeed, for each x,

$$z^{m(x)} = \prod_{i=1}^{n} (1 + (z-1) f_i(x)) = \sum_{I \subseteq [n]} (z-1)^{|I|} \prod_{i \in I} f_i(x).$$

Sum over x and group the subsets I by size.

Exactly r properties. Taking the coefficient of z^r in (11) gives

$$\boxed{N_r = \sum_{k=r}^{n} (-1)^{k-r} \binom{k}{r} S_k \qquad (0 \le r \le n).} \tag{12}$$

For instance, $N_0 = S_0 - S_1 + S_2 - \cdots$ and $N_1 = S_1 - 2S_2 + 3S_3 - \cdots$. The forward relation is

$$S_k = \sum_{r=k}^{n} \binom{r}{k} N_r. \tag{13}$$

It follows either by counting the pairs above or by substituting $z = 1 + t$ in (11) and comparing coefficients of t^k.

At least r properties. For $1 \le r \le n$,

$$\boxed{N_{\ge r} := \sum_{j=r}^{n} N_j = \sum_{k=r}^{n} (-1)^{k-r} \binom{k-1}{r-1} S_k.} \tag{14}$$

Sum (12) over $j \ge r$ and use $\sum_{j=r}^{k} (-1)^{k-j} \binom{k}{j} = (-1)^{k-r} \binom{k-1}{r-1}$, which follows from Pascal's identity. The case $r = 1$ is ordinary PIE. Also, $N_{\le r} = |X| - N_{\ge r+1}$ for $0 \le r < n$.

When intersections depend only on their size. If $|A_I| = a_k$ whenever $|I| = k$, then $S_k = \binom{n}{k} a_k$. For example, $N_0 = \sum_{k=0}^{n} (-1)^k \binom{n}{k} a_k$. This reduces a sum over 2^n subsets to $n + 1$ terms.

An odd number of properties (symmetric difference). Since $(P(1) - P(-1))/2$ counts objects of odd multiplicity,

$$|A_1 \triangle \cdots \triangle A_n| = \sum_{k=1}^{n} (-2)^{k-1} S_k. \tag{15}$$

Here $\triangle$ denotes symmetric difference. The even-multiplicity count is $|X|$ minus this number.

8. Probability and Bonferroni bounds

Probability versions. For events $A_1, \dots, A_n$ in any probability space, put

$$s_0 = 1, \qquad s_k = \sum_{\substack{I \subseteq [n] \\ |I|=k}} \Pr(A_I), \qquad M = \sum_{i=1}^{n} \mathbf{1}_{A_i}.$$

Taking expectations of the finite indicator identities gives

$$\Pr(M = 0) = \sum_{k=0}^{n} (-1)^k s_k, \qquad \Pr(M = r) = \sum_{k=r}^{n} (-1)^{k-r} \binom{k}{r} s_k. \tag{16}$$

No independence assumption is needed. The full distribution is encoded by

$$\mathbb{E}[z^M] = \sum_{k=0}^{n} s_k (z-1)^k, \qquad s_k = \mathbb{E}\binom{M}{k}.$$

Thus intersection sums are the binomial moments of the number of events that occur. Mutual independence gives the additional factorization $\mathbb{E}[z^M] = \prod_{i=1}^{n} (1 - p_i + p_i z)$, where $p_i = \Pr(A_i)$.

Bonferroni inequalities. Return to counting, with the S_k of the preceding section, and set $B_\ell = \sum_{k=1}^{\ell} (-1)^{k-1} S_k$ for $1 \le \ell \le n$. Then

$$\begin{cases} B_\ell \le |\bigcup_i A_i|, & \ell \text{ even}, \\ B_\ell \ge |\bigcup_i A_i|, & \ell \text{ odd}. \end{cases} \tag{17}$$

For example,

$$S_1 - S_2 \;\le\; \left| \bigcup_i A_i \right| \;\le\; S_1.$$

These bounds are useful when only low-order intersections are known.

Proof. An object lying in exactly $j \ge 1$ sets contributes to B_ℓ the sum

$$\sum_{k=1}^{\ell} (-1)^{k-1} \binom{j}{k} = 1 - (-1)^{\ell} \binom{j-1}{\ell}.$$

The last binomial coefficient is nonnegative, so the contribution is at most 1 for even ℓ and at least 1 for odd ℓ. An object in no set contributes 0. Summing proves the bounds. Here $\binom{a}{b} = 0$ for integers $a \ge 0$ and $b > a$.

Bounds for avoiding all properties. If $C_\ell = \sum_{k=0}^{\ell} (-1)^k S_k$, then $C_\ell \le N_0$ for odd ℓ and $C_\ell \ge N_0$ for even ℓ. In particular,

$$S_0 - S_1 \;\le\; N_0 \;\le\; S_0 - S_1 + S_2.$$

The same bounds hold for probabilities after replacing S_k by s_k, and for nonnegative weighted counts. They follow by subtracting (17) from the size of the universe.

9. Subset inversion and binomial inversion

Möbius inversion on subsets. For functions f, g on the subsets of $[n]$, with values in an abelian group,

$$g(S) = \sum_{T \subseteq S} f(T) \quad \Longleftrightarrow \quad f(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} g(T). \tag{18}$$

There is also an upper-set form:

$$g(S) = \sum_{T \supseteq S} f(T) \quad \Longleftrightarrow \quad f(S) = \sum_{T \supseteq S} (-1)^{|T|-|S|} g(T). \tag{19}$$

These formulas recover exact data from cumulative data.

Proof of (18)**.** Substitute the first formula into the second. The coefficient of $f(U)$ is

$$\sum_{U \subseteq T \subseteq S} (-1)^{|S|-|T|} = \sum_{j=0}^{|S \setminus U|} (-1)^j \binom{|S \setminus U|}{j} = \begin{cases} 1, & U = S, \\ 0, & U \neq S. \end{cases}$$

Thus only $f(S)$ remains. The reverse implication follows by the same calculation. Reversing inclusions proves the upper-set form.

How ordinary PIE fits. Let $f(T)$ count the objects belonging to exactly the sets indexed by T, and let $g(S) = |A_S|$. Then $g(S) = \sum_{T \supseteq S} f(T)$. Taking $S = \varnothing$ in (19) gives the avoidance formula. Taking an arbitrary S gives the exact-membership formula (8) with $Q = [n] \setminus S$.

Ordinary binomial inversion. For two sequences (a_j) and (b_j),

$$b_m = \sum_{j=0}^{m} \binom{m}{j} a_j \quad \Longleftrightarrow \quad a_m = \sum_{j=0}^{m} (-1)^{m-j} \binom{m}{j} b_j. \tag{20}$$

This is subset inversion when the value on a subset depends only on its cardinality. Equations (13) and (12) give the corresponding finite upper-triangular version.

The general finite-poset version. If P is a finite partially ordered set, define $\mu(u, u) = 1$ and, for $u < v$, $\mu(u, v) = -\sum_{u \leq t < v} \mu(u, t)$. Then

$$g(v) = \sum_{u \leq v} f(u) \quad \Longleftrightarrow \quad f(v) = \sum_{u \leq v} \mu(u, v) g(u).$$

The matrix with entries $\mathbf{1}_{u \leq v}$, in a linear extension of the order, is upper triangular with diagonal 1. The defining recurrence says that $(\mu(u, v))$ is its inverse, which proves the formula. For subsets, $\mu(U, V) = (-1)^{|V \setminus U|}$, recovering PIE.

10. Four standard applications

1. Derangements and prescribed numbers of fixed points. For permutations of $[n]$, let $A_i = \{\pi : \pi(i) = i\}$. Fixing a specified set of k points leaves $(n-k)!$ permutations. Consequently,

$$D_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

More generally, the number with exactly r fixed points is $\binom{n}{r} D_{n-r}$: choose the fixed points and derange the rest. This also follows directly from (12).

2. Surjections. For functions $[m] \to [n]$, where $m, n \geq 1$, let A_i mean that the value i is missing from the image. For $|I| = k$, there are $(n-k)^m$ functions missing every value in I. Hence

$$\#\{\text{surjections } [m] \to [n]\} = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m.$$

Exactly r missing values can instead be counted using (12).

3. Compositions with upper bounds. Let $m \geq 0$ and $b_1, \ldots, b_n \geq 0$ be integers, with $n \geq 1$. To count solutions of $x_1 + \cdots + x_n = m$ with $0 \leq x_i \leq b_i$, declare $A_i = \{x_i \geq b_i + 1\}$. Shifting these coordinates gives

$$\#\{\text{solutions}\} = \sum_{I \subseteq [n]} (-1)^{|I|} \binom{m - \sum_{i \in I} (b_i + 1) + n - 1}{n - 1}.$$

Here the term is defined to be 0 whenever $m - \sum_{i \in I}(b_i + 1) < 0$. Otherwise it is the usual stars-and-bars count. This convention is necessary, even when the upper entry of the binomial coefficient is negative.

4. Coprimality and the arithmetic sieve. For integers $N, q \geq 1$, let the forbidden properties be divisibility by the distinct prime factors of q. Intersections correspond to divisibility by products of those primes. Thus

$$\#\{1 \leq a \leq N : \gcd(a, q) = 1\} = \sum_{d \mid q} \mu(d) \left\lfloor \frac{N}{d} \right\rfloor.$$

Here $\mu(1) = 1$, $\mu(d) = (-1)^k$ if d is a product of k distinct primes, and $\mu(d) = 0$ if a prime square divides d. For $N = q$, this yields $\varphi(q) = q \prod_{p \mid q} (1 - 1/p)$. The related divisor-inversion formula is

$$G(q) = \sum_{d \mid q} F(d) \quad \Longleftrightarrow \quad F(q) = \sum_{d \mid q} \mu(d)\, G(q/d).$$

It follows by substitution and $\sum_{d \mid q} \mu(d) = \mathbf{1}_{\{q=1\}}$, itself an expansion of $\prod_{p \mid q} (1 - 1)$ when $q > 1$.

Further reading and a guide to the formulas

What is being counted?	Useful form
Objects having at least one property	Ordinary PIE, (5)
Objects avoiding all forbidden properties	Avoidance, (7)
Required and forbidden properties together	Prescribed membership, (8)
An intersection, given union counts	Dual PIE, (9)
Total weights or generating functions	Weighted PIE, (10)
Objects having exactly r properties	Exact sieve formula, (12)
Objects having at least r properties	Threshold formula, (14)
The whole multiplicity distribution	Sieve polynomial, (11)
An odd number of properties	Symmetric difference, (15)
Dependent events in a probability space	Probability PIE, (16)
Bounds from a few intersection sums	Bonferroni, (17)
Exact data from cumulative subset data	Subset inversion, (18)–(19)
Sequences related by binomial sums	Binomial inversion, (20)

References. The proofs above are self-contained. The following sources provide further examples and broader perspectives.

1. Jiří Matoušek and Jaroslav Nešetřil, *Invitation to Discrete Mathematics*. A reference for the basic inclusion–exclusion principle and elementary counting.

2. Ira M. Gessel and Richard P. Stanley, *Algebraic Enumeration*, Sections 12–13: inclusion–exclusion and Möbius inversion. Author-hosted text.

3. *Probability Models and Statistics*, Chapter 8, lecture notes hosted by the Indian Institute of Science: inclusion–exclusion and Bonferroni inequalities. Online notes.

4. NIST, *Digital Library of Mathematical Functions*, Section 27.5, *Inversion Formulas*. Divisor Möbius inversion.